

Handling Missing Data: A Comparative Study of Imputation Strategies

Missing data is the first consequential decision in any machine learning pipeline, and the wrong imputation choice silently corrupts every model built afterwards. This project demonstrates and compares five mainstream strategies for handling missing values — column dropping, mode imputation, median imputation, arbitrary-value imputation, and KNN imputation — alongside categorical encoding methods, using the classic Titanic dataset of 891 passengers as a realistic testbed: 177 Age values are missing (19.87%), the Cabin field is 77.10% empty (687 records), and Embarked has 2 gaps.

Each strategy is matched to the situation it suits. Cabin, missing beyond any honest recovery threshold at 77%, is dropped entirely — imputing at that level manufactures data rather than restoring it. Embarked, a categorical field with a dominant class, is filled with its mode, Southampton (raising its count from 644 to 646). Age receives the most instructive treatment: its distribution shows meaningful outliers (maximum 80 years against a median of 28, with a mean of 29.7 pulled upward by the tail), so the median is deliberately chosen over the mean — a decision justified visually via boxplot before imputing, not assumed.

The project then makes the cost of careless imputation visible. Arbitrary values (9999 and -1) are injected into Age and the resulting distributions overlaid against the original using kernel density plots — showing the artificial spikes such values create, which can usefully flag “missingness” to tree-based models but severely mislead distance-based algorithms. As the higher-fidelity alternative, KNNImputer with five nearest neighbours fills 133 missing training-set ages by borrowing from similar passengers, preserving the distribution's natural shape.

Throughout, pipeline discipline is maintained: a 75/25 train–test split (668 training, 223 test records) is performed before imputation so fill values are learned exclusively from training data, preventing leakage. The encoding companion notebook completes the preprocessing toolkit, contrasting One-Hot Encoding (with reference categories dropped to avoid the dummy-variable trap) against Label Encoding implemented both via scikit-learn and pandas category codes.

The result is a practical decision guide — drop above ~70% missing, mode for sparse categorical gaps, median for skewed numerics, KNN when relationships matter, and always split before imputing — turning a routine preprocessing step into a reasoned, defensible methodology.